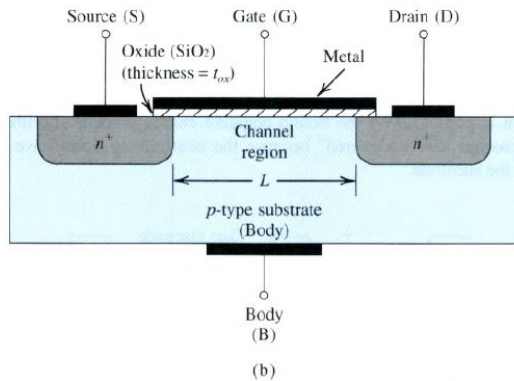
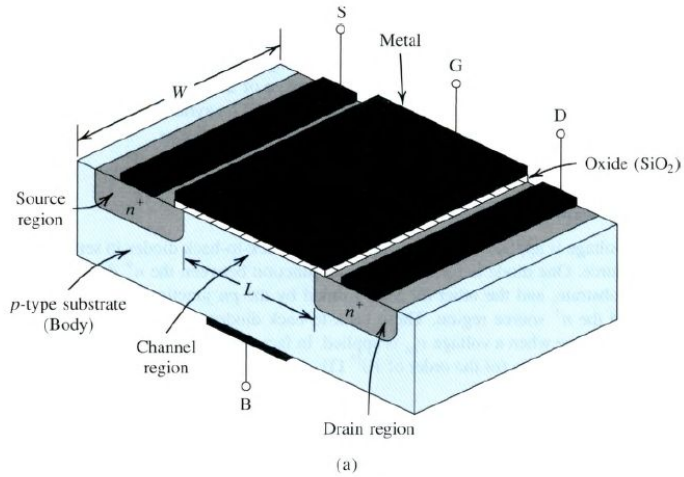


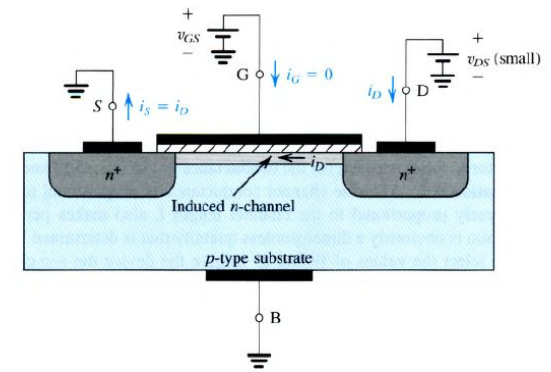
CSE350: Digital Electronics & Pulse Techniques

Lecture 13: MOSFET Basics

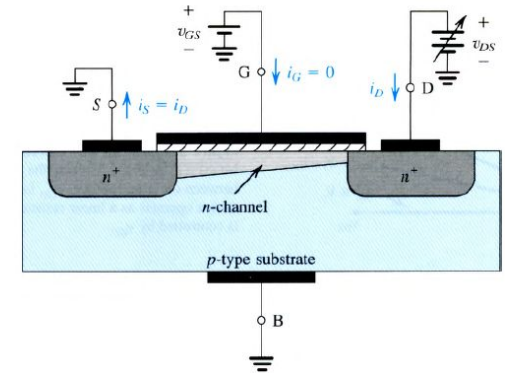
NMOS



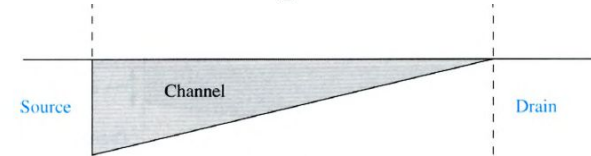
$$v_{DS} = 0, v_{GS} > v_{TN}$$



$$0 < v_{DS} < v_{OV} = v_{GS} - v_{TN}$$



$$v_{DS} = v_{OV} = v_{GS} - v_{TN}$$



I-V Characteristics of NMOS:

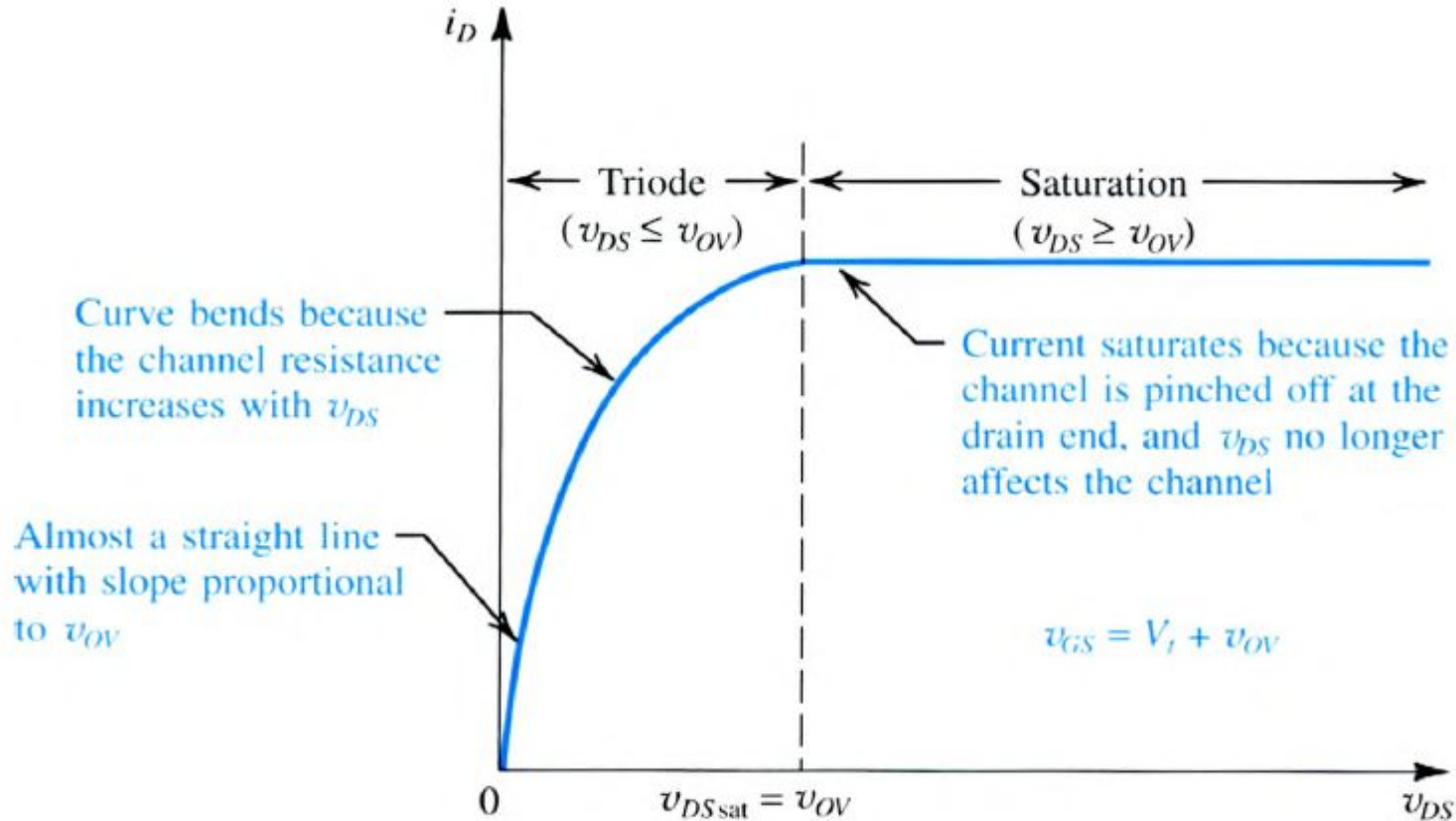


Table 3.1

Summary of the MOSFET current–voltage relationships

NMOSNonsaturation region ($v_{DS} < v_{DS}(\text{sat})$)

$$i_D = K_n[2(v_{GS} - V_{TN})v_{DS} - v_{DS}^2]$$

Saturation region ($v_{DS} > v_{DS}(\text{sat})$)

$$i_D = K_n(v_{GS} - V_{TN})^2$$

Transition point

$$v_{DS}(\text{sat}) = v_{GS} - V_{TN}$$

Enhancement mode

$$V_{TN} > 0$$

Depletion mode

$$V_{TN} < 0$$

PMOSNonsaturation region ($v_{SD} < v_{SD}(\text{sat})$)

$$i_D = K_p[2(v_{SG} + V_{TP})v_{SD} - v_{SD}^2]$$

Saturation region ($v_{SD} > v_{SD}(\text{sat})$)

$$i_D = K_p(v_{SG} + V_{TP})^2$$

Transition point

$$v_{SD}(\text{sat}) = v_{SG} + V_{TP}$$

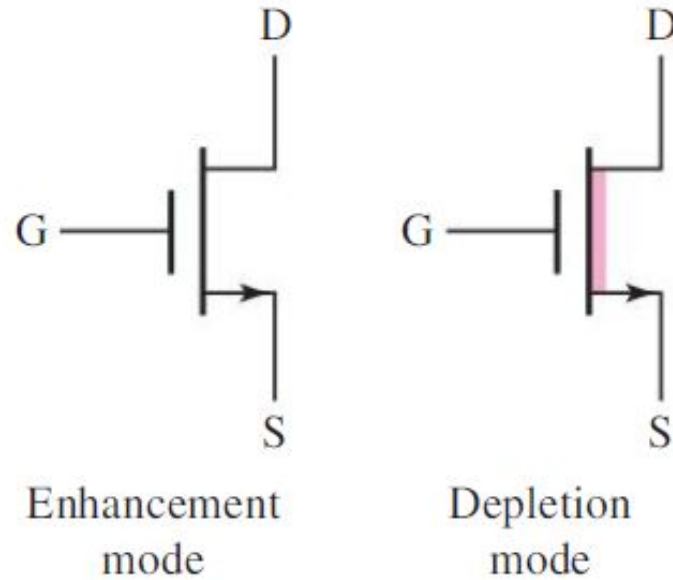
Enhancement mode

$$V_{TP} < 0$$

Depletion mode

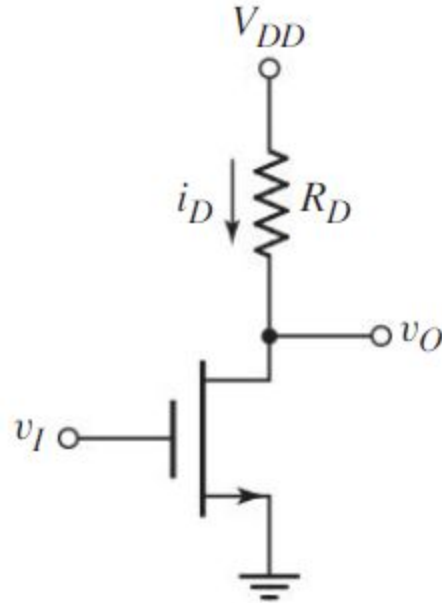
$$V_{TP} > 0$$

NMOS Symbols



(a)

NMOS Inverter:



Example 1: For the NMOS inverter, determine V_o for $V_I = 1.5V$ & $5V$. It is given that,

$$V_{DD} = 5V, R_D = 20k\Omega, V_{TN} = 0.5V, K_n = 0.3mA/V^2$$

Solution:

$$V_{GS} = V_I = 5V$$

Since the circuit will act like an inverter, the higher the input voltage, the smaller the output voltage,

$$V_o = V_{DD} - I_{DS}R_D$$

So, we may assume that the NMOS is in triode region for this input voltage, i.e.

$$V_{DS} = V_o < V_{GS} - V_{TN} = 5 - 0.5 = 4.5V$$

For triode region, we know,

$$I_{DS} = K_n[2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2] = 0.3[2 * 4.5 * V_{DS} - V_{DS}^2]$$

Again,

$$I_{DS} = \frac{V_{DD} - V_o}{R_D} = \frac{5 - V_o}{20}$$

$$\text{Thus, } 0.3[9V_{DS} - V_{DS}^2] = \frac{5 - V_o}{20}$$

$$\text{Solving, } V_o = 9.074V, 0.092V$$

But 9.074V will make drain current negative, so not possible. So, $V_o = 0.092V$

Verification: Since, V_o or V_{DS} is less than overdrive voltage (4.5V), then our assumption of triode region operation is correct.

$$V_{GS} = V_I = 1.5V$$

Since the voltage is quite small, we may assume that the output voltage will be high (since it's an inverter).

Thus, it can be that, $V_o = V_{DS} > V_{GS} - V_{TN}$, indicating saturation mode operation. For saturation, we know, $I_{DS} = K_n[(V_{GS} - V_{TN})]^2 = 0.3[1.5 - 5]^2 = 0.3mA$

$$\text{Now, } I_{DS} = \frac{V_{DD} - V_o}{R_D} = \frac{5 - V_o}{20} = 0.3$$

$$\text{Solving, } V_o = -1V$$

Verification: Since, output voltage is less than overdrive voltage ($1.5 - 0.5 = 1V$), it cannot be in saturation. Thus, our assumption was wrong. So, the NMOS must be in triode region. A similar analysis to the previous case assuming triode region operation gives, $V_o = 1.667V, 0.5V$

Now, the 1.667V is not acceptable, since it is greater than overdrive voltage (1V). Thus the output must be 0.5V, which aligns with our assumption of triode region.

Verification: Output voltage is 0.5V, which is less than the overdrive voltage. Thus, our assumption is correct.

Example 2: Determine the transition point, maximum drain current, and maximum power dissipation of an NMOS inverter with resistor. (The same inverter as the previous problem, just $V_{DD} = 2.5V$)

Solution:

Observation: When output voltage has the smallest value, then, $i_D = \frac{V_{DD} - V_o}{R_D}$ would have a maximum value. Therefore maximum input voltage is needed to produce this output value.

We can take the maximum value of input voltage at 2.5V. So,

$$V_I = 2.5V \text{ [We may assume, } V_{DS} = V_{DD} - I_D R_D < V_{DD} = V_G = V_I \text{ since the } I_D \text{ is max]}$$

Thus, the NMOS will be operating in the triode region.

$$\text{So, } I_D = \frac{2.5 - V_o}{20} = 0.3[2(2.5 - 0.5)V_o - V_o^2]$$

$$\text{Or, } 6V_o^2 - 25V_o + 2.5 = 0$$

Solving, $V_o = 0.1025V, 4.064V$. $4.064V$ is not acceptable, since it would mean the current is flowing in the opposite direction. Thus, $V_o = 0.1025V < 2.5 - 0.5 = 2 = V_{GS} - V_{TN}$.

So, our assumption was valid.

$$\text{Now, } I_{D_{max}} = \frac{2.5 - 0.1025}{20} = 0.1198 \text{ mA}$$

Maximum power dissipation,

$$P_{max} = \Delta V I_{D_{max}} = (V_{DD} - 0) I_{D_{max}} = (2.5 - 0) * 0.1025 = 0.3 \text{ mW}$$

Transition point:

$$\text{At transition point, } V_{DS} = V_o = V_{GS} - V_{TN} = V_I - V_{TN} \text{ \& } I_D = K_n (V_{GS} - V_{TN})^2$$

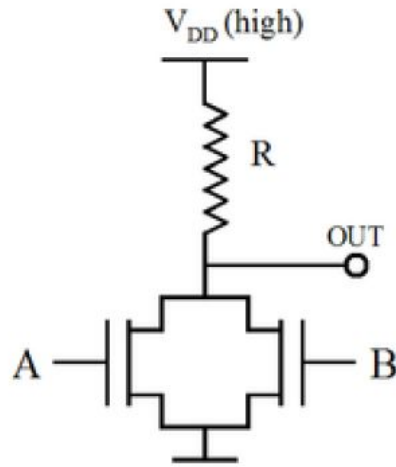
$$\text{So, } I_D = \frac{V_{DD} - V_o}{R_D} = K_n (V_{GS} - V_{TN})^2 = K_n V_o^2$$

$$\text{Or, } \frac{2.5 - V_o}{20} = 0.3 V_o^2$$

$$\text{Or, } 6V_o^2 + V_o - 2.5 = 0$$

Solving, $V_o = 3.4051V$, $0.5675V$. $3.4051V$ is not acceptable for the same reason. Thus, at transition point, $V_o = 0.5675V$, $V_I = V_o + V_{TN} = 1.0675V$

NMOS NOR Gate Example



Example 3: Determine the current & voltages in this gate for different input conditions:

$$K_n = 0.1mA/V^2, R = 20k\Omega, V_{TN} = 0.8V$$

Solution:

Case (0,0):

$V_A = V_B = 0V$. For NMOS A, we have, $V_{GSA} = V_{GA} - V_{SA} = 0 - 0 = 0 < 0.8V$

Similarly, $V_{GSB} = V_{GB} - V_{SB} = 0 - 0 = 0 < 0.8V$. Thus, both the NMOS are in cutoff mode. Hence, $I_{DA} = I_{DB} = I_D = 0V$, $V_o = V_{DD} = 5V$

Case (0,1|1,0):

$V_A = 0V$, $V_B = 5V$. Thus, $V_{GSA} = 0 - 0 = 0 < 0.8$, $V_{GSB} = 5 - 0 = 5 > 0.8V$

Hence, NMOS A is in cutoff, while NMOS B can be assumed to operate in triode region (since there might be some drop across the resistance and drain voltage will then be less than 5V. Also, since it is a NOR gate, the expected output voltage is low for these cases)

So, the full current from the source will flow through NMOS B. We get,

$$I_D = I_{DB} = \frac{V_{DD} - V_o}{20} = K_{nB} [2(V_{GSB} - V_{TNB})V_{DSB} - V_{DSB}^2]$$

$$\text{Or, } \frac{5 - V_o}{20} = 0.1 [2(5 - 0.8)V_o - V_o^2]$$

Solving, $V_o = 8.6V, 0.29V$. But $8.6V$ output indicates negative current, hence not acceptable. Thus, the output voltage for these cases is $0.29V$, which is lower than overdrive voltage ($5-0.8=4.2V$), thus our assumption of triode mode is correct.

$$\text{Now, } I_D = I_{DB} = \frac{5-0.29}{20} = 0.235 \text{ mA}$$

Case (1,1):

$$V_A = V_B = 5V; V_{GSA} = 5 - 0 = 5 > 0.8, V_{GSB} = 5 - 0 = 5 > 0.8$$

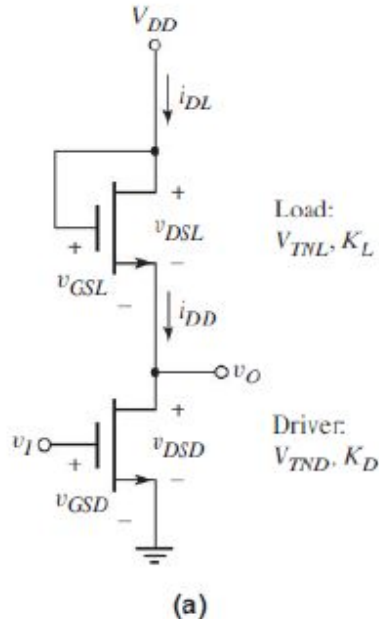
We can again assume that both the NMOS are operating in triode mode, for the same reasons. Now,

$$\begin{aligned} I_D = I_{DA} + I_{DB} &= \frac{5-V_o}{20} = K_{nA}[2(V_{GSA} - V_{TNA})V_o - V_o^2] + K_{nB}[2(V_{GSB} - V_{TNB})V_o - V_o^2] \\ &= 2 * 0.1[2(5 - 0.8)V_o - V_o^2] \end{aligned}$$

Solving, $V_o = 4.17V, 0.149V$. Now, $4.17V$ is much closer to $4.2V$, that would drive the NMOS to saturation. So, the correct value of output voltage is $0.149V$, which also aligns with the assumption of triode region operation.

$$\text{So, } I_D = \frac{5-0.149}{20} = 0.243 \text{ mA}; I_{DA} = I_{DB} = \frac{I_D}{2} = 0.121 \text{ mA}$$

NMOS Inverter with Enhancement Load:



Example 4: Determine the DC currents & voltages in the circuit containing enhancement load.

$$V_{TND} = V_{TNL} = 1V, K_{nD} = 50 \mu A/V^2$$

$$K_{nL} = 10 \mu A/V^2$$

Determine output voltage for input of 5V & 1.5V.

Solution:

Case (1.5V):

Observation regarding load NMOS: We can see that, $V_{GL} = V_{DL} = 5V$. Therefore,

$$V_{DSL} = V_{DL} - V_{SL} = V_{GL} - V_{SL}$$

$$\text{Thus, } V_{DL} - V_{SL} \geq (V_{GL} - V_{SL}) - V_{TNL} = V_{GSL} - V_{TNL}$$

So, load NMOS is always in saturation, regardless of input & output voltage.

For driver NMOS, we can assume saturation mode too (since we are expecting output voltage high compared to $V_{GSD} - V_{TND}$).

$$\text{Using KCL, } I_{DL} = I_{DD} \rightarrow K_{nL}(V_{GSL} - V_{TNL})^2 = K_{nD}(V_{GSD} - V_{TND})^2$$

$$\rightarrow 0.01(5 - V_o - 1)^2 = 0.05(1.5 - 0 - 1)^2$$

$$\rightarrow V_o = 2.88V > V_{GSD} - V_{TND}$$

$$\text{Thus, our assumption is valid. Now, } I_{DL} = I_{DD} = 0.05(1.5 - 1)^2 = 1.25 * 10^{-2} \text{ mA}$$

Case(5V):

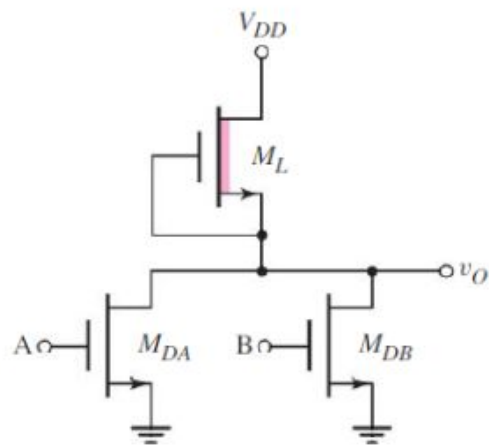
In this case, since the input is high, we are expecting low output voltage. Thus we may assume that the driver NMOS is in triode mode. The load is again in saturation, for the reason stated in the previous case.

$$\text{Again, KCL gives, } I_{DL} = I_{DD} \rightarrow K_{nL}(V_{GSL} - V_{TNL})^2 = K_{nD}[2(V_{GSD} - V_{TND})V_o - V_o^2]$$
$$\rightarrow 0.01(5 - V_o - 1)^2 = 0.05[2(5 - 0 - 1)V_o - V_o^2] \rightarrow V_o = 7.65V, 0.34V$$

Since $7.65V > 5V$, the current would be negative. Thus, the output voltage is $0.34V$, which is also lower than the overdrive voltage ($5-1 = 4V$). Thus, our assumption is valid.

$$\text{So, } I_{DL} = I_{DD} = 0.01(5 - 0.34 - 1)^2 = 0.133 \text{ mA}$$

NMOS NOR Gate with Depletion Load:



Example 5: Determine the output voltage of the NMOS NOR gate.

$$V_{DD} = 2.5V, K'_n = 100 \mu A/V^2, V_{TND} = V_{TNA} = V_{TNB} = 0.4V$$
$$V_{TNL} = -0.6V, \left(\frac{W}{L}\right)_D = 4, \left(\frac{W}{L}\right)_L = 1$$

Note: $K_n = K'_n \left(\frac{W}{L}\right)$

Solution:

Case(1,1): Observation: As this is a NOR gate, when two inputs are high, output voltage should be low. Therefore, V_o can be assumed to be low in such cases.

For the load depletion type NMOS, we have, $V_{DSL} = V_{DL} - V_{SL} = 2.5 - V_o \rightarrow$ this value is large.

Thus,

$V_{DSL} \geq V_{GSL} - V_{TNL} = 0 - (-0.6) = 0.6 \rightarrow 5 - V_o \geq 0.6$. Hence, for this case, we can deduce that the load NMOS will be in saturation mode.

For the two driver NMOS A & B, we are expecting them to operate in triode region, due to the small voltage at their drains.

Now, KCL at output gives, $I_{DL} = I_{DA} + I_{DB}$

$$\rightarrow K_{nL}(V_{GSL} - V_{TNL})^2 = K_{nA}[2(V_{GSA} - V_{TNA})V_o - V_o^2] + K_{nB}[2(V_{GSB} - V_{TNB})V_o - V_o^2]$$

$$\rightarrow K_{nL}'\left(\frac{W}{L}\right)_L(0 - (-0.6))^2 = K_{nA}'\left(\frac{W}{L}\right)_D[2(2.5 - 0.4)V_o - V_o^2] + K_{nB}'\left(\frac{W}{L}\right)_D[2(2.5 - 0.4)V_o - V_o^2]$$

$$\rightarrow V_o = 4.178V, 0.02153V$$

4.178V is greater than the supply voltage and would drive the current in the opposite direction. Thus, not acceptable. So, the output voltage is 0.02153V. Now,

$$I_{DL} = K_{nL}'\left(\frac{W}{L}\right)_L(V_{GSL} - V_{TNL})^2 = 0.01 * 1(0 - (-0.6))^2 = 0.036 \text{ mA}$$

$$I_{DA} = K_{nA} \left(\frac{W}{L} \right)_D [2(V_{GSA} - V_{TNA})V_o - V_o^2] = 0.01 * 4[2(2.5 - 0.4)0.02153 - 0.02153^2]$$

$$= 0.018 \text{ mA}$$

$$I_{DB} = I_{DL} - I_{DA} = 0.018 \text{ mA}$$

The other cases can be solved in a similar fashion, and left as an exercise.